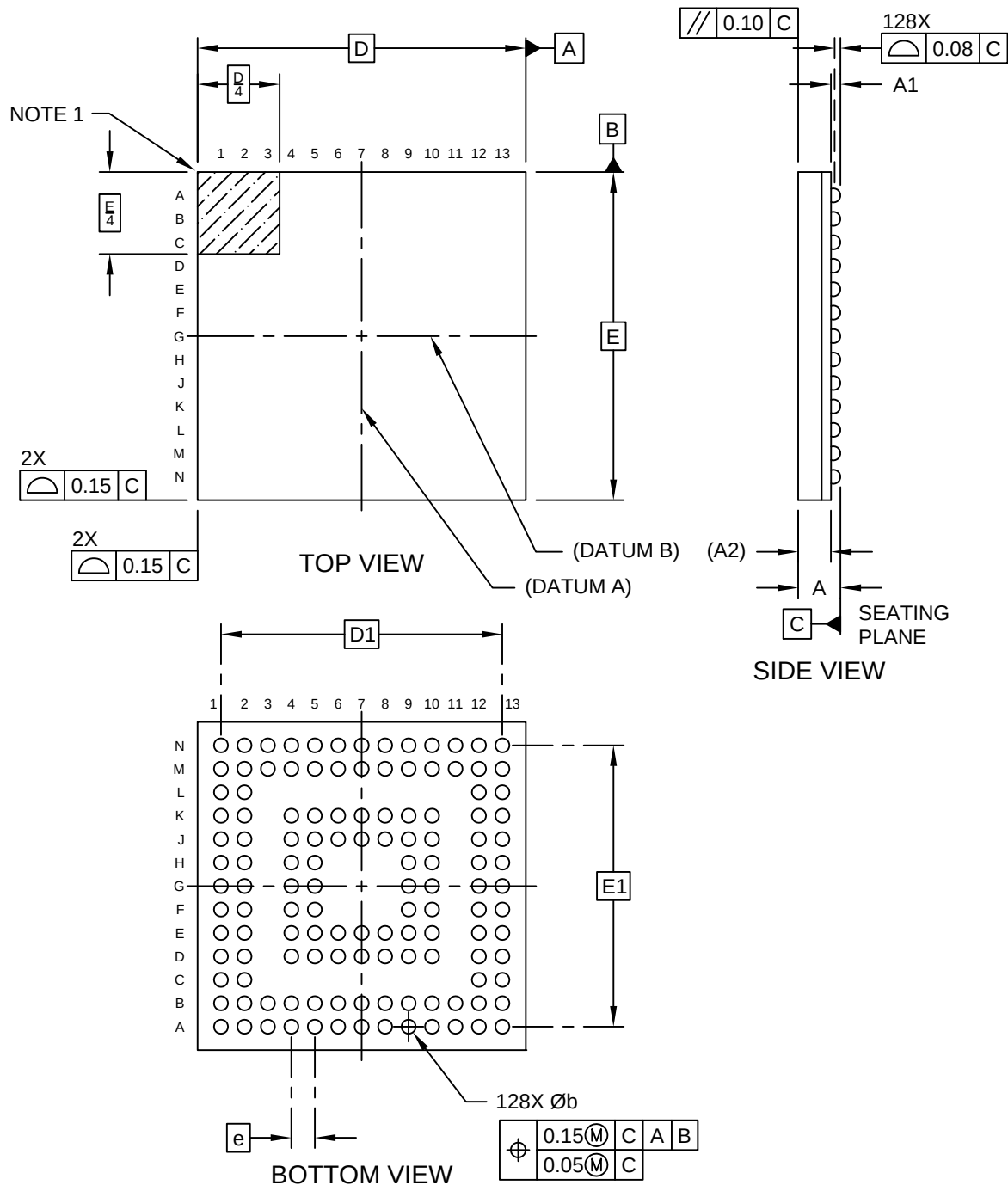


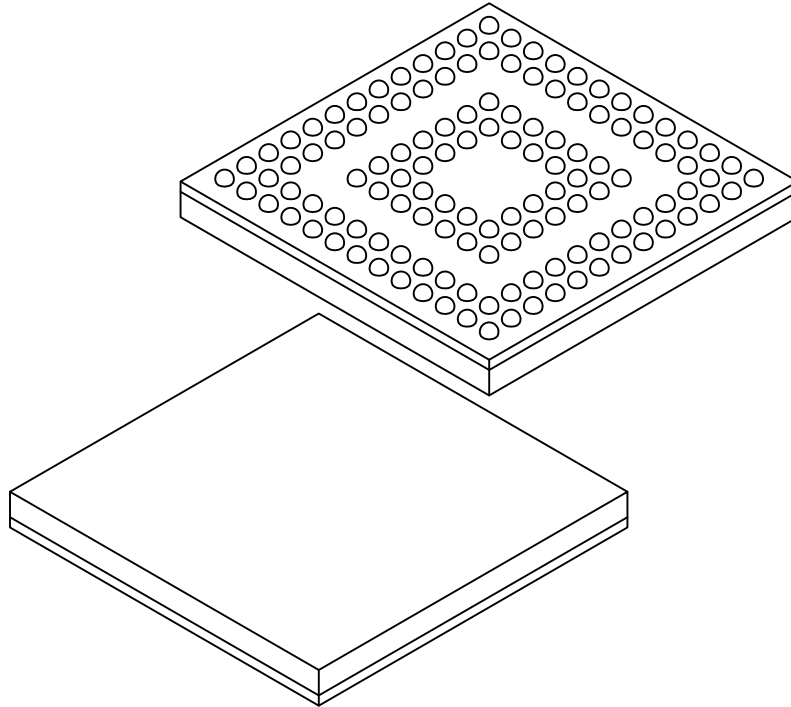
128-Ball Very, Very Thin Fine Pitch Ball Grid Array (TFX) - 7x7 mm Body [WFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



128-Ball Very, Very Thin Fine Pitch Ball Grid Array (TFX) - 7x7 mm Body [WFBGA]

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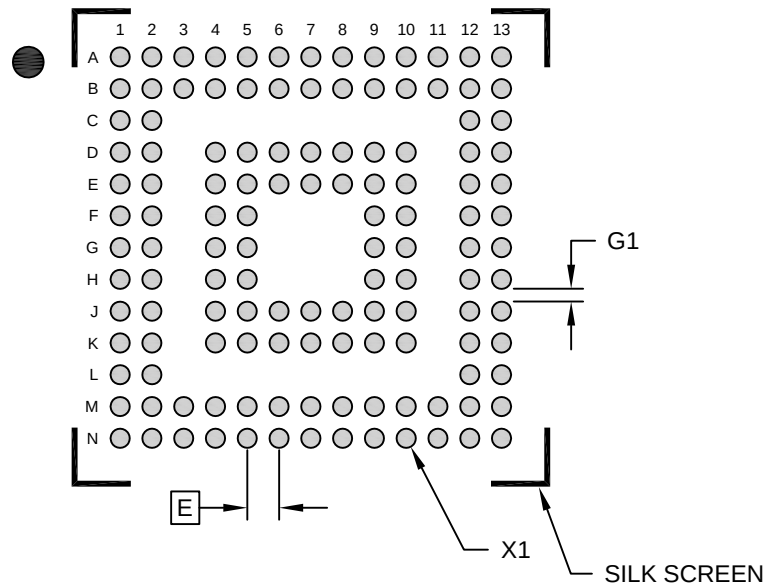
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	128		
Pitch	e	0.50 BSC		
Overall Height	A	-	0.70	0.80
Standoff	A1	0.12	0.17	0.22
Molded Package Thickness	A2	0.53 REF		
Overall Length	D	7.00 BSC		
Overall Pitch	D1	6.00 BSC		
Overall Width	E	7.00 BSC		
Overall Pitch	E1	6.00 BSC		
Ball Diameter	b	0.20	-	0.30

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

128-Ball Very, Very Thin Fine Pitch Ball Grid Array (TFX) - 7x7 mm Body [WFBGA]

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RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Contact Pad Diameter (X128)	X1			0.32
Contact Pad to Contact Pad	G1	0.20		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process